COMPUTER PROGRAMMING • PRACTICE

Multiplication Table

DESCRIPTION

Consider an $n \times n$ multiplication table: the number in row i and column j ($1 \le i, j \le n$) is $i \cdot j$. For example, when $n = 3$ the table is

```
1  2  3
2  4  6
3  6  9
```

Suppose all n^2 numbers of the table are written down in non-decreasing order. Since n is odd, the list has an odd length, and its middle element is the $((n^2 + 1) / 2)$-th smallest number. Find this middle element.

For $n = 3$ the sorted list is 1, 2, 2, 3, 3, 4, 6, 6, 9, so the answer is 3.

INPUT FORMAT

The input consists of a single line containing an odd integer n.

OUTPUT FORMAT

Print one integer: the middle element of the sorted list.

CONSTRAINTS

$1 \le n < 10^6$, and n is odd.

SAMPLE

Sample 1 Input

```
3
```

Sample 1 Output

```
3
```

SOURCE

Adapted from the CSES Problem Set, task 2422.